

Hybridizing Metaheuristics with Generative AI Techniques for Selected Graph Problems

Research project for Bosnia and Herzegovina–Austria bilateral cooperation

1. Background and motivation

Combinatorial optimization (CO) is a cornerstone of modern decision support in logistics, energy systems, telecommunication networks, transportation, and data-driven planning. Graph problems such as maximum independent set (MIS), MaxCut, densest k-subgraph extraction, minimum consistent subset selection, and dynamic electric autonomous dial-a-ride problems can all be formulated as large-scale combinatorial search tasks in which the quality of the solution depends on complex structural constraints, uncertainty, and interactions between decisions. Classical exact methods provide guarantees but often become intractable on large or dynamic instances, while traditional metaheuristics can scale better but rely heavily on hand-crafted operators, parameter tuning, and domain-specific engineering (Blum and Roli, 2003; Glover and Kochenberger, 2003; Toth and Vigo, 2002).

The central challenge is not merely to find a feasible solution, but to discover high-quality, diverse, and robust search trajectories in very large combinatorial spaces. This is exactly where machine learning and generative AI can be strategically valuable. Data-driven models can identify structural patterns, approximate promising regions of the search space, and support decisions such as which vertices to include, which parts of the solution to destroy, which components to merge, and how to control the search process adaptively. Recent work in learning-enhanced optimization and hybrid metaheuristics has shown promising gains in component selection, branch-and-bound guidance, repair operators, and constructive heuristics (Lodi and Prouvost, 2022; Mandi et al., 2024; Reixach and Blum, 2024; Reixach and Blum, 2025).

The proposed project builds on these developments and extends them by integrating generative models, including diffusion models and generative flow networks (GFlowNets), into classical hybrid optimization frameworks such as large neighborhood search (LNS), Construct, Merge, Solve and Adapt (CMSA), and column generation (CG). The project hypothesis is that implicit knowledge encoded by graph-aware neural models and generative architectures can guide the search more effectively than purely hand-designed operators, especially when combined with exact repair or incumbent-improving subproblems. This is particularly relevant for graph formulations where structure matters: e.g., a good destroy set in LNS, a promising candidate component in CMSA, or a policy for deciding when to re-optimize or to repair a partial solution. Recent research on diffusion-based evolutionary search, GFlowNets for destroy sets, and deep-learning enhancements of CMSA provides an excellent scientific foundation for this bilateral project (Akbay et al., 2025; Bresich et al., 2025; Bresich et al., 2026; Soler and Raidl, 2025; Wüstinger et al., 2026).

The project is designed as a bilateral initiative between the partner institutions in Austria and Bosnia and Herzegovina. It will combine Austrian expertise in algorithmic optimization, machine learning, and hybrid search with Bosnian expertise in combinatorial optimization, learning-based search, and applied algorithmic design. The research will be organized around a common methodological core, a set of benchmark graph problems, and two applied domains with strong

societal and industrial relevance: consistent subset selection and dynamic electric autonomous dial-a-road problem (EADARP) variants.

2. State of the art and research gap

The field of learning-enhanced optimization has developed rapidly. Methodological surveys show that machine learning can improve branching, candidate generation, heuristic policies, and algorithm configuration for exact and approximate solvers (Lodi and Prouvost, 2022; Mandi et al., 2024). Within metaheuristics, data-driven control has proven useful for neighborhood selection, operator scheduling, solution construction, and evolutionary search. In this context, CMSA and LNS are particularly attractive because the quality of their solutions depends strongly on how promising partial solutions, components, and destroy sets are created and selected (Reixach and Blum, 2024; Reixach and Blum, 2025; Akbay et al., 2025).

Recent advances in generative modeling provide a more expressive alternative to standard learned heuristics. GFlowNets are designed to sample structured objects with probability proportional to reward, which makes them natural tools for generating diverse candidate subsets and destroy sets in solving graph problems (Bengio et al., 2022; Bengio et al., 2023; Bresich et al., 2026). Diffusion models, originally successful in image generation, have also been adapted to combinatorial optimization, where they can generate promising graph structures, improve evolutionary search, or provide candidate solutions for hard graph problems (Soler and Raidl, 2025; Wüstinger et al., 2026). These methods are especially relevant because graph optimization frequently requires not only a single optimal solution but a distribution of good, diverse candidate structures; this diversity is crucial in LNS, CMSA, and hybrid repair pipelines.

Decision-focused learning (DFL) is another emerging direction. Rather than minimizing prediction error alone, DFL trains models to reduce the regret of downstream optimization decisions. This is particularly relevant in dynamic routing and resource allocation settings, where a prediction error may be harmless if it does not affect the decision quality, and harmful if it leads to bad operational choices (Mandi et al., 2024; Baty et al., 2024). The dynamic EADARP is a natural test bed because it involves future demand, charging, and time-window constraints that necessitate anticipatory decisions rather than myopic choices. Learning to schedule LNS neighborhoods, decide which request to replan, or predict future feasibility can significantly improve practical performance (Tomandl et al., 2026; Bresich et al., 2025; Bresich et al., 2026).

Despite these advances, a clear research gap remains. The literature still lacks a unified framework that tightly integrates structured search control, generative models, and classical decomposition methods in one optimization loop. In particular, most current methods either (i) rely on end-to-end neural solvers that perform poorly on hard graph instances when compared to hybrid methods, or (ii) use machine learning only as a local add-on to a heuristic without end-to-end search control. Moreover, references to LLM-based heuristic generation remain largely exploratory and are not yet embedded into a rigorous optimization pipeline for graph problems. The project addresses precisely this gap by developing a hybrid search architecture that combines classical optimization, graph-aware learning, generative models, and decision-focused control in one coherent framework.

3. Scientific objectives

The project pursues the following objectives:

- (i) To design graph-aware learning mechanisms for component selection, destroy-set generation, and constructive heuristics in hybrid optimization methods such as CMSA and LNS.
- (ii) To integrate diffusion models, GFlowNets, and related generative approaches into hybrid search procedures for graph optimization, with emphasis on diversity, quality, and multi-objective performance.
- (iii) To develop decision-focused learning strategies for dynamic routing and scheduling applications, especially in the context of EADARP and related online decision problems.
- (iv) To investigate the use of large language models (LLMs) as generators of lower-level heuristics, scoring functions, and search-control rules that can be tested and integrated within a reproducible algorithmic framework.
- (v) To validate these methods on benchmark graph problems and applied decision scenarios including the minimum consistent subset problem, maximum independent set, MaxCut, dense k-subgraph problems, and dynamic EADARP.

The overall goal is to create a hybrid algorithmic pipeline that fuses the robustness of exact subroutines and metaheuristics with the expressiveness and generalization power of modern generative and neural techniques.

4. Proposed methodology and work packages

The project will be organized around a set of interconnected work packages that mirror the repository structure and the scientific agenda of the proposal.

WP1. Hybrid LNS–CMSA–column generation framework

This work package will develop a generic optimization engine that combines LNS, CMSA, and column generation. The framework will treat constructive processes, component generation, and repair subproblems as learnable modules rather than fixed algorithmic ingredients. We will generate promising partial solutions and candidate components through graph-aware heuristics and then integrate them using exact or hybrid subsolvers. This package will provide a strong methodological backbone for all application domains and a modular architecture for plugging in learned policies or generative models (Akbar et al., 2025; Reixach and Blum, 2024; Reixach and Blum, 2025).

WP2. Diffusion models and GFlowNets for destroy-set and candidate generation

The second work package will focus on generative architectures for combinatorial structures. Diffusion models and GFlowNets will be used to generate structured destroy sets, candidate subsets, and component pools that are more informative than randomly sampled neighborhoods.

For LNS, the learned model will propose destroy sets that target critical vertices, edges, or components capable of triggering strong repair phases. For CMSA, the model will generate candidate components or partial solutions whose inclusion is likely to improve the objective value. The key research challenge is to combine diversity, feasibility, and reward-aware generation in a way that is compatible with exact repair and search intensification. This direction is aligned with recent work on denoising diffusion-based search and GFlowNets for structure generation (Bengio et al., 2022; Bengio et al., 2023; Soler and Raidl, 2025; Wüstinger et al., 2026; Bresich et al., 2026).

WP3. Generative AI (LLMs) for lower-level heuristic synthesis

This work package will explore the use of large language models as algorithmic design assistants. LLMs will be used to generate candidate lower-level heuristics, repair operators, scoring functions, neighborhood definitions, and control policies from a high-level specification. These generated routines will be translated into executable code, automatically filtered by feasibility and performance checks, and integrated into the optimization loop. The goal is not to replace the solver, but to create a structured human-AI co-design framework in which LLMs propose candidate algorithmic ingredients and the optimizer evaluates them under real-performance criteria. This package is especially relevant for the robust and adaptive generation of hybrid heuristics for graph-based optimization. Related work on LLM-directed algorithm discovery and optimization-guided heuristics motivates this direction (Yang et al., 2023; Liu et al., 2024).

WP5. Application 1: Minimum consistent subset problem

The first application will focus on the minimum consistent subset problem (MCSP) and related variant formulations. The objective is to identify a small subset of constraints, observations, or graph elements that preserves the essential structure of a larger system while enabling faster optimization or more interpretable decision making. This problem is relevant to feature selection, sparse structure recovery, and decision support in graph-based inference. The project will benchmark exact formulations, hybrid decomposition methods, and learned component-generation strategies on synthetic and real-world instances. We will compare heuristic quality, speed, and interpretability against baseline methods.

WP6. Application 2: (Dynamic) electric autonomous dial-a-ride problem

The second application is the dynamic EADARP. This problem includes pickup and delivery requests, fleet constraints, charging station availability, time windows, battery constraints, and dynamic changes in demand. It is a difficult real-world scenario in which learning-based anticipatory decisions are particularly valuable. The project will use static EADARP instances to calibrate the hybrid framework and then extend it to dynamic variants in which the solver must decide when to re-optimize, which requests to postpone, and how to adapt charging decisions online. This package is designed to evaluate the practical impact of the proposed machine learning and generative components in a realistic logistics setting (Bresich et al., 2025; Tomandl et al., 2026; Bresich et al., 2026).

5. Expected results and impact

The project will generate several tangible outputs. First, it will produce a unified hybrid optimization framework that combines classical search with graph-aware learning, generative models, and decision-focused control. Second, it will deliver a benchmark suite of graph instances and application scenarios covering MIS, MaxCut, densest k-subgraph, MCSP, and dynamic EADARP variants. Third, it will create reusable algorithmic modules for destroy generation, candidate construction, and heuristic policy control. Fourth, it will produce open and reproducible software components in a repository structure aligned with the planned work packages.

The expected scientific impact is substantial. The project will advance the state of the art by showing how generative AI and learning can be integrated into robust optimization rather than used as isolated end-to-end models. It will also generate concrete methodological knowledge for the Austrian and Bosnian teams, including joint publications, thesis topics, student mobility, and transferable software for industrial optimization. The project is expected to yield both methodological advances and practical contributions to logistics, infrastructure planning, and large-scale graph decision making.

6. References

- [1] Y. Bengio, E. J. Hu, and colleagues, Generative flow networks: a new way to sample structured objects, NeurIPS 2022.
- [2] Y. Bengio, S. Lahlou, T. Deleu, Y. B. Ji, and others, GFlowNets and variational inference for structured generation, arXiv:2023.
- [3] C. Blum and A. Roli, Metaheuristics in combinatorial optimization: overview and conceptual comparisons, ACM Computing Surveys, 2003.
- [4] L. Baty, K. Jungel, P. S. Klein, A. Parmentier, and M. Schiffer, Combinatorial optimization-enriched machine learning to solve the dynamic vehicle routing problem with time windows, Transportation Science, 2024.
- [5] M. A. Akbay, C. Blum, and G. R. Raidl, A hybrid CMSA-column generation approach to variable-sized bin packing, ECAI 2025.
- [6] M. Bresich, G. R. Raidl, and S. Limmer, Revisiting large neighborhood search with on-the-fly charging station insertion for the electric autonomous dial-a-ride problem, 2025.
- [7] M. Bresich, J. Peng, G. R. Raidl, and S. Limmer, Determining destroy sets in large neighborhood search by generative flow networks, 2026.
- [8] M. Bresich, G. R. Raidl, C. Coleman, P. Welke, and S. Limmer, Search space reduction through machine learning for the electric autonomous dial-a-ride problem, 2026.

- [9] J. Reixach and C. Blum, How to improve Construct, Merge, Solve and Adapt? Use reinforcement learning!, *Annals of Operations Research*, 2024.
- [10] J. Reixach and C. Blum, Improving the CMSA algorithm with online deep learning, *ECAI 2025*.
- [11] A. Lodi and A. Prouvost, Learning to optimize: a survey of modern approaches, *Operations Research*, 2022.
- [12] J. Mandi, A. G. G. de Sousa, F. D. H. de Witte, T. M. B. ..., and others, Decision-focused learning: foundations, state of the art, benchmark and future opportunities, *Journal of Artificial Intelligence Research*, 2024.
- [13] X. Liu, Y. Wang, and others, LLM-driven optimization and algorithm synthesis, 2024.
- [14] F. Glover and G. Kochenberger, *Handbook of Metaheuristics*, Springer, 2003.
- [15] P. Toth and D. Vigo, *The Vehicle Routing Problem*, SIAM, 2002.
- [16] J. Salvà Soler and G. R. Raidl, A denoising diffusion-based evolutionary algorithm framework: application to the maximum independent set problem, *arXiv:2510.08627v1*, 2025.
- [17] M. Wüstinger, E. Iurlano, and G. R. Raidl, A denoising diffusion adaptive search for the α -domination problem on social graphs, 2026.
- [18] L. Tomandl, M. Bresich, G. R. Raidl, Y. Mei, S. Limmer, and T. Rodemann, A reinforcement learning guided large neighborhood search for the dynamic electric autonomous dial-a-ride problem, 2026.
- [19] F. Odelga, E. Iurlano, and G. R. Raidl, A Construct Merge Solve and Adapt heuristic for least cost influence maximization, 2026 (submitted)
- [20] J. Yang, X. Wang, and others, Large language models as optimizers, *arXiv preprint*, 2023.